

Introduction to Image Analysis

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1 Assignment Description

2 Code Limitations

2.1 C++

3 Assignment Details

3.1 Image Loading

3.1.1 Image Investigations

3.1.2 Image loading using `libtiff`

3.1.2.1 Loading *tiled* and *striped* formats

3.1.2.1.1 Anomalies encountered

3.1.2.1.2 Anomaly description and reasoning

3.1.2.2 Loading and visualizing sub-images of the `.svs` file

3.1.2.3 Loading subregions of images

3.1.3 Image combination

3.2 Intensity Transformations

3.2.1 Power law for intensities (disregarding constant c)

3.2.2 Piece-wise linear intensity transformations

3.2.3 Histogram calculation for grayscale images

3.2.4 Histogram equalization

3.2.5 Lookup table intensity transformations

3.3 Spatial Filtering and Multistep Image Processing

3.3.1 Image classes for pixel intensities

3.3.1.1 Methods of conversion

3.3.2 Image blurring by spatial filtering

3.3.3 Numerical approximations of derivative operators

3.3.4 Image sharpening using an intermediate Laplacian calculation

3.3.4.1 Benefits

3.3.4.2 Drawbacks

3.3.5 Multistep image processing

3.4 Image Transformations

3.4.1 Calculation of index-to-local and local-to-world transformations

3.4.2 Composite affine transformations

3.4.3 Nearest Neighbor and Bi-linear Interpolation for visualization of transformed images

3.4.4 Visualization of affine transformations

3.4.4.1 Nearest Neighbor type

3.4.4.2 Bi-linear type

3.4.5 QA of AI classification of kidney histology images